

Math 526, F09
Quiz 2

Name: Answer
SSN: xxx-xx-

Instructions: There are totally 2 problem. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (10 points)

(a) Compute $\mathbf{Ax}$ for the matrix $\mathbf{A}$ and the vector $\mathbf{x}$ using dot products of the rows of $\mathbf{A}$ with the column vector $\mathbf{x}$ where

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 4 \\ -2 & 3 & 1 \\ -4 & 1 & 2 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix}.$$

Solution:

$$\mathbf{Ax} = \begin{bmatrix} (1, 2, 4) \cdot (2, 2, 3) \\ (-2, 3, 1) \cdot (2, 2, 3) \\ (-4, 1, 2) \cdot (2, 2, 3) \end{bmatrix} = \begin{bmatrix} 2+4+12 \\ -4+6+3 \\ -8+2+6 \end{bmatrix} \\ = \begin{bmatrix} 18 \\ 5 \\ 0 \end{bmatrix}$$

(b) Multiply these two matrices:

$$\begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 0 \\ 1 & 3 & 1 \end{bmatrix}.$$

Solution:

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & -6 \\ 0 & 1 & -2 \end{bmatrix}$$

Problem 1. (10 points) Write down the 3×3 matrices that produce the following operations:

(a) E_{21} subtracts 5 times row 1 from row 2.

$$E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -5 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(b) E_{32} adds 7 times row 2 to row 3.

$$E_{32} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 7 & 1 \end{bmatrix}$$